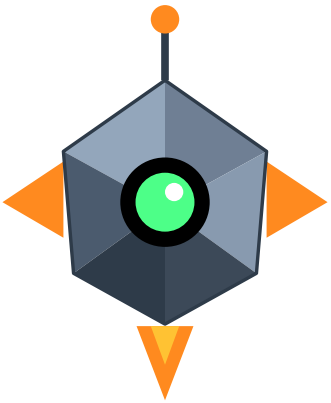
BOOK 9

Solving by Changing the Question

Changing the question, keeping the answer





MULTIX // HELLO
I am Multix. I ride along in every chapter. Watch my readout: it shows what is owed, what is paid, and what is only a guess.

CHAPTER 1

Ask It Another Way



MULTIX // CIRCLE CHECK
A circle holds a piece that belongs together, and circles can sit inside circles. A circle is *ready* when everything needed for its calculation is known and its operation is allowed by the current rules. To **pop** a ready circle, replace it with its value and leave everything outside it alone. To **redraw**, move circles around or split a number, as long as the value stays the same. The full rules are in the introduction.

STORY

Pia had a habit that annoyed her older brother. Whenever he asked her a hard question, she answered a *different* question, and somehow that gave the right answer.
"What's 1003 minus 997?" he asked, hoping to trip her up.
"How far is it from 997 to 1003?" said Pia. "Six."
She was right. She had not dodged the question. She had asked it another way.

STEP 1 Subtraction as distance

Subtracting 997 from 1003 looks awkward, with all that borrowing. But picture a number line.

997 is 3 below 1,000. 1,003 is 3 above 1,000. The distance from one to the other is 3 + 3 = **6**.

The question "what is 1003 − 997?" turned into "how far apart are they?" Both ask for the same number, and the second is easy.

Here is another one: **83 − 29**. Instead of subtracting, count *up* from 29 to 83.

- 29 to 30: 1
- 30 to 80: 50
- 80 to 83: 3

1 + 50 + 3 = **54**. Check by subtracting: 83 − 29 is 54.

IN CIRCLES

Pia's move is a redraw. She changes where the circles are, so that every circle is easy to pop.

Trust check. A change of question is trustworthy when you can say **what it preserves** and **what it drops**, and when you have **checked the answer in the original question**.

TRACE

```
( 1003 − 997 )
redraw  →  ( ( 1000 − 997 ) + ( 1003 − 1000 ) )
pop ( 1000 − 997 )  →  3
pop ( 1003 − 1000 )  →  3
( 3 + 3 )
pop  →  6
```

The redraw is allowed because it keeps the value: going from 997 up to 1,003 is the same as going up to 1,000 and then on to 1,003. Whenever you change the question, ask whether the redraw keeps the value, and what it forgets.

STEP 2 Multiplying as a rectangle

Now a product: **13 × 14**. Pia imagined a rectangle 13 wide and 14 tall, then cut each side into a 10 and a leftover:

- 13 is 10 + 3
- 14 is 10 + 4

The big rectangle breaks into four smaller ones:

- 10 × 10 = 100
- 10 × 4 = 40
- 3 × 10 = 30
- 3 × 4 = 12

100 + 40 + 30 + 12 = **182**.

The question “what is 13 × 14?” became “what is the area of this rectangle?”, and the area was easy to add up from four easy pieces.

STEP 3 An equation as a hot-and-cold game

Here is an equation: **x + 7 = 12**.

Pia changed the question to a game of hot and cold. She made a **gap**: the left side minus the right side.

gap = (x + 7) − 12

The goal is to find the value of x that makes the gap **zero**. A gap of zero means the two sides are equal.

- Try x = 3. Gap: (3 + 7) − 12 = −2. Too low.
- Try x = 4. Gap: (4 + 7) − 12 = −1. Closer.
- Try x = 5. Gap: (5 + 7) − 12 = **0**. Found it!

In this puzzle, the sign of the gap tells you which way to go: a negative gap means the left side is too small, so try a bigger

x. (That is true for $x + 7 = 12$. In other equations a bigger x can push the gap the other way, so always look.)

STEP 4 What did the new question keep, and what did it lose?

Changing the question is powerful, but there is a catch. A new question is only as good as what it *keeps*. Before Pia trusts a change, she asks two things:

1. **What does the new question keep?**
2. **What might it lose, or change?**

Here is what each of her moves kept and lost.

Original	New version	Kept	Lost or changed
$29 - 83$	Distance between 29 and 83	How big the difference is (54)	Which one is bigger. The subtraction was −54, but the distance is 54.
13×14	A rectangle with its sides labelled	The total area, 182, and both sides	If you write down only the area, which sides made it. A 12 by 1 rectangle and a 3 by 4 rectangle have the same area.
$x + 7 = 12$	Gap equals zero	The solution, $x = 5$	Nothing, as long as the gap is exactly left side minus right side
12 cookies shared among 5 kids, whole cookies only	$12 \div 5 = 2.4$	The amount per kid	The whole-cookie rule. Nobody can get 2.4 cookies.

The last row is the one to watch. The division gave a perfectly good number, but the number did not fit the original question.

The real answer is that each kid gets **2** cookies, with **2** left over.

THE RULE: TAKE THE ANSWER HOME

Pia has a rule she never breaks: **always take the answer back to the original question and ask whether it fits.**

- Did 6 really equal 1003 − 997? Yes.
- Did x = 5 really make x + 7 equal 12? Yes.
- Did 2.4 cookies each really fit "whole cookies only"? No, so go back and fix it.

Changing the question is a way to find an answer. Checking against the original question is how you know your answer fits. (It does not tell you that no other answers exist. That needs its own reason.)

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did Pia keep when she changed 1003 − 997 into a distance?** (The answer, 6.)
- **What can a changed question lose?** (Details of the original, like direction, or a rule such as "whole cookies only.")
- **What does Pia always do at the end?** (Take the answer back to the original question and check that it fits.)

WHAT MAKES THIS CHANGE TRUSTWORTHY?

A change of question is trustworthy when you can say **what it preserves** and **what it drops**, and when you have **checked the answer in the original question**. That is a stronger standard than "it gave an answer." Anything can give an answer. What matters is that the answer still belongs to the question you started with.

PRACTICE

1. Find **1005 − 995** by asking how far apart the numbers are.
2. Find **72 − 38** by counting up from 38.
3. Find **12 × 15** by splitting it into four rectangles.
4. Solve **x + 9 = 21** using the gap. What do you get

for the gap when x = 10? When x = 12?

5. Twelve cookies are shared among 5 kids, whole cookies only. What does the division 12 ÷ 5 = 2.4 lose?

6. **29 − 83** is a negative number, but the distance from 29 to 83 is positive. What did the distance version lose?

7. Redraw **52 − 48** as two gaps through 50, and pop every circle.

BRIDGE *Bridge to the rest of the series:* In Reversing the Calculation, the same area can come from different rectangles, as 12 can be 3 × 4 or 2 × 6, so an area on its own forgets how it was made. In The Choice Before the Calculation, rewriting a problem is a choice, and here every choice comes with a list of what it keeps. In A Useful Wrong Answer, checking at a far mark and checking in the original question are the same habit: take the answer home and see whether it fits.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × 3 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × 3 )  →  9
6 pop ( 1,600 − 9 )  →  1,591
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × ( 1 + 1 ) ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × ( 1 + 1 ) )  →  2
6 pop ( 400 − 2 )  →  398
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × 6 ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × 6 )  →  42
6 pop ( 4,900 − 42 )  →  4,858
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 2

The Change That Added Too Much

STORY

Pia loved changing questions. One day she tried a new move on the equation **x + 1 = 5**.
"I'll multiply both sides by 0," she said. "It will be 0 on the left and 0 on the right."

TRACE

```
( x + 1 ) × 0  =  5 × 0
0  =  0
```

"Zero equals zero," said Pia. "That is true! So the answer is… every number?"
Her older brother laughed. "Check it in the original."

STEP 1 Take the answer home

Pia picked x = 7 and took it back to the original question:

TRACE

```
x + 1 = 5   with x = 7:   7 + 1 = 8,  and 8 is not 5.
```

So x = 7 is **not** an answer to the original. But it *was* an answer to the new question (0 = 0 is true for every x). The new question had added answers that the old one did not have.

This is a cousin of the lesson from chapter 1. There, a change could lose something. Here, a change can **gain** something it should not.

STEP 2 Why did multiplying by 0 add answers?

When you multiply both sides by 0, you throw away the information on both sides. Everything turns into 0. Before the change, the equation was fussy: only x = 4 worked. After, nothing is fussy: every x works.

Multiplying both sides by 2 is different:

Trust check. A change of question is trustworthy when you can undo it, because then it keeps exactly the same answers.

TRACE

```
( x + 1 ) × 2  =  5 × 2
2x + 2  =  10
```

You can also *undo* it, by dividing both sides by 2, and you are back at x + 1 = 5. Because the change can be undone, no answers were added or lost.

That gives a rule:

A change keeps the answers if you can undo it.

Multiplying by 2 can be undone (divide by 2). Adding 3 to both sides can be undone (subtract 3). Multiplying by 0 cannot be undone, because every number times 0 is 0. You met this in Reversing the Calculation: a button with no undo forgets where it came from.

STEP 3 A change that can lose an answer

Here is a puzzle: **x × 3 = x × 5**.

Pia first tried dividing both sides by x:

TRACE

```
3  =  5
```

“That is false,” she said. “So there are no answers.”

But then she tried x = 0 in the original:

TRACE

```
0 × 3 = 0   and   0 × 5 = 0
```

Both sides are 0! So x = 0 **is** an answer. The new question 3 = 5 had lost it.

Why? Dividing by x cannot be undone when x is 0. You cannot divide by 0. So the change was only safe for starting values other than 0. Pia learned to say: “I divided by x, so I need to check x = 0 separately.”

STEP 4 In circles

Here is Step 3 as a picture. Each side is a circle with a mystery x inside.

TRACE

```
( x × 3 )  =  ( x × 5 )
```

Neither circle is ready, since each holds x. Pia can test a value by dropping it in, which makes the circles ready:

TRACE

```
x = 0:   ( 0 × 3 ) = ( 0 × 5 )   pop both  →  0 = 0   ✓
x = 2:   ( 2 × 3 ) = ( 2 × 5 )   pop both  →  6 = 10  ✗
```

Dropping a value in and popping is the test that never changes the question. It always checks against the original.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did multiplying by 2 keep?** (Exactly the same answers.)
- **What did multiplying by 0 do?** (It added answers that were not in the original.)
- **What did dividing by x do?** (It lost the answer x = 0.)

WHAT MAKES THIS TRUSTWORTHY?

A change of question is trustworthy when you can undo it, because then it keeps exactly the same answers. If it cannot be undone, then either it added answers, or it lost some, and you have to check.
The check that always works is to take the answer home, as Pia's brother said. That catches **added** answers. It does not catch **lost** answers, because you cannot test a number you never found. To catch lost answers, ask: "Is there any value where my change would not be allowed?"

PRACTICE

1. Solve x + 2 = 9 by adding 3 to both sides, then solving. Can the change be undone?
2. Pia changes x + 2 = 9 by multiplying both sides by 0. Which numbers now make the new question true? Which of them are answers to the original?
3. Check x = 0 and x = 4 in the puzzle x × 6 = x × 6. What do you notice?
4. Pia divides both sides of x × 2 = x × 7 by x. What does she get? Is there an answer she lost?
5. A button doubles a number. Another button multiplies it by 0. Which one could you use as a safe change of question? Why?

BRIDGE *Bridge to the rest of the series:* In chapter 1, Pia checked that her answer fit the original question. Here that check catches an added answer, but only a second question, “where is my change not allowed?”, catches a lost one. In Reversing the Calculation, the same test, can it be undone, tells you whether a button forgets.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,496**. Leave the others blank. What appears?

2476 + 21	2496 + 8	2495 + 6	2502 − 6	2512 − 16	2460 + 29	2492 + 11	2468 + 26
2500 + 6	2496 + 5	2488 + 18	2523 − 27	2473 + 23	2487 + 5	2469 + 21	2487 + 17
2477 + 28	64 × 64 − 40 × 40	2491 + 5	2527 − 31	2527 − 31	32 × 78	2510 − 14	2476 + 16
2516 − 20	2535 − 39	2470 + 19	2469 + 27	2528 − 32	2477 + 7	52 × 48	64 × 39
2506 − 10	2482 + 14	2491 + 5	2474 + 22	2506 − 10	2487 + 9	2471 + 25	16 × 156
2 × 1248	2509 − 13	2514 − 18	2468 + 28	32 × 78	2509 − 13	2504 − 8	2485 + 11
2494 + 11	2528 − 32	2466 + 30	2491 + 12	2468 + 29	2493 + 3	2533 − 37	2472 + 19
2482 + 11	2501 − 5	2481 + 15	20 × 125	2460 + 28	2521 − 25	2498 − 2	15 × 166



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 3

Slide Both Numbers

STORY

Pia noticed that her bus fare and her brother's bus fare were different, but each year both fares went up by the same amount, so the *gap* between them stayed the same. The gap did not care about the shift.
That gave Pia an idea for subtraction.

STEP 1 Slide both numbers by the same amount

Take **83 − 29**. Imagine both numbers on a number line. What if you slide both of them one step to the right?

TRACE

```
( 83 − 29 )  →  54
( 84 − 30 )  →  54
```

The gap between them did not change. And 84 − 30 is much easier, because 30 is a round number.

This is a redraw that changes the question without changing the answer:

TRACE

```
( 83 − 29 )
redraw  →  ( 84 − 30 )
pop  →  54
```

More examples:

- **612 − 498.** Slide both up by 2: 614 − 500 = **114**.
- **1,003 − 997.** Slide both up by 3: 1,006 − 1,000 = **6**.
- **91 − 48.** Slide both up by 2: 93 − 50 = **43**.

STEP 2 Why it works

Subtraction asks, "how far apart?" If you move both numbers the same distance in the same direction, the distance between them stays exactly the same.

You can also *undo* the slide: slide both back. A change that can be undone, as in chapter 2, keeps the answers.

Trust check. A change is trustworthy when you know what it keeps.

STEP 3 A warning, and a new move for sums

Sliding both numbers does **not** work for adding.

TRACE

```
( 29 + 83 )  →  112
( 30 + 84 )  →  114
```

The answers are different. Adding does not ask about the gap, it asks about the total, and sliding both numbers up adds the slide twice.

But there is a similar move for sums: **move one number up and the other down by the same amount**.

TRACE

```
( 29 + 83 )
redraw  →  ( 30 + 82 )
pop  →  112
```

What one number gained, the other lost, so the total is the same. Another example: 198 + 47 → (200 + 45) → **245**.

So the rule depends on what the question asks.

- A **difference** stays the same when both numbers slide **together**.
- A **sum** stays the same when one slides **up** and the other slides **down**.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What does sliding both numbers keep for subtraction?** (The gap between them, which is the answer.)
- **What does it lose?** (The numbers themselves. 84 − 30 no longer mentions 83 and 29.)

WHAT MAKES THIS TRUSTWORTHY?

A change is trustworthy when you know what it keeps. Subtraction keeps the gap, so a slide together is safe. A sum keeps the total, so a swap of up and down is safe. The wrong move for the wrong question, like sliding both numbers for a sum, gives a wrong answer even though it looks similar.
As in chapter 1, always take the answer home. Check 54 in the original: 83 − 29 = 54, and 29 + 54 = 83. ✓

PRACTICE

1. Find 502 − 297 by sliding both numbers up by 3.
2. Find 91 − 48 by sliding both up by 2. Check your answer by adding 48 to it.
3. Find 6,001 − 2,998.
4. Find 396 + 127 by moving 4 from one number to the other.
5. Dani says: “52 + 38 is the same as 54 + 40, since I slid both up by 2.” What is wrong with that?
6. Redraw (71 − 39) by sliding both numbers and pop it.

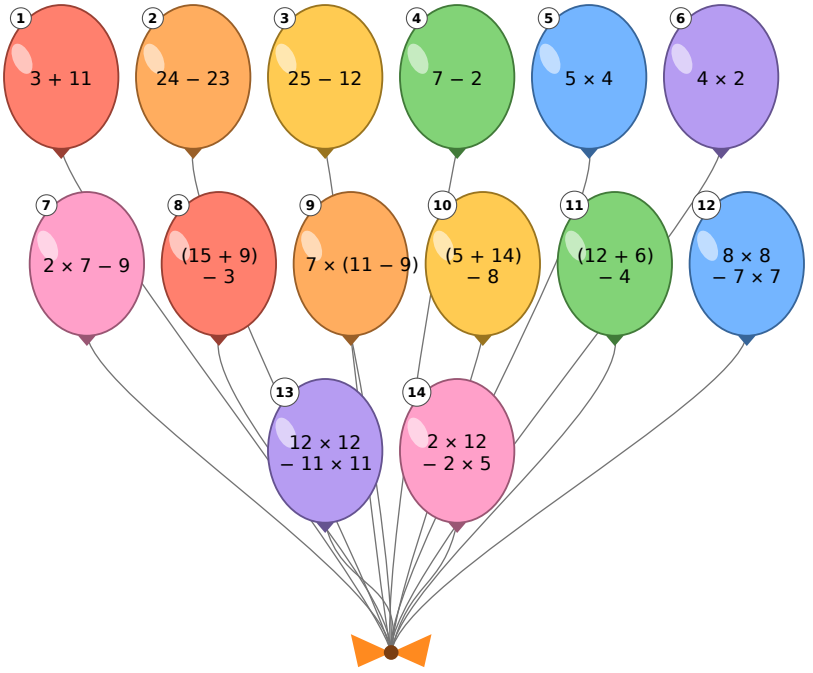
BRIDGE *Bridge to the rest of the series:* In chapter 2, a change was safe if it could be undone. A slide can be undone by sliding back, which is why it is safe for subtraction. In The Work That Vanishes, (8 × 12) became (10 × 10) − (2 × 2) by moving one number up and the other down, which is the same kind of swap as in Step 3.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

CHAPTER 4

Scale Both

STORY

Pia was given a hard division: **360 ÷ 15**. She did not want to do it. Her brother said, “You can change the question.”
“By how much?”
“Divide both numbers by the same thing.”

STEP 1 Shrink both numbers

Pia divided both 360 and 15 by 5.

TRACE

```
( 360 ÷ 15 )
redraw  →  ( 72 ÷ 3 )
pop  →  24
```

And 72 ÷ 3 is much easier. Check in the original: 15 × 24 = 360. ✓

The answer to a division does not change when both numbers are scaled the same way. Think of it as sharing: 360 cookies among 15 children gives each the same amount as 72 cookies among 3 children, because everything is smaller by the same factor.

Another one: **1,200 ÷ 25**. Multiply both by 4:

TRACE

```
( 1,200 ÷ 25 )
redraw  →  ( 4,800 ÷ 100 )
pop  →  48
```

Check: 25 × 48 = 1,200. ✓

STEP 2 What this does not do to subtraction

Subtraction behaves differently. Double both numbers in **100 − 40**:

TRACE

```
( 100 − 40 )  →  60
( 200 − 80 )  →  120
```

Trust check. The trust comes from knowing what the new question keeps and what it changes.

The answer also doubled. So scaling both numbers keeps a **division** answer but doubles a **subtraction** answer. You would need to halve it again to get back.

What each move keeps depends on what the question asks.

STEP 3 A catch with leftovers

Now take a division with a leftover: **17 ÷ 5**. It is 3 with 2 left over.

Double both numbers:

TRACE

```
( 34 ÷ 10 )  →  3 with 4 left over
```

The 3 stayed the same. The leftover did not: it went from 2 to 4.

Why? Because the leftover is measured in pieces of the same size as the divisor. Two out of 5 is the same share as four out of 10. The leftover doubled because the pieces were twice the size.

So the scaling trick keeps the quotient, but the leftover must be scaled back if you want the original.

Another: **33 ÷ 6** is 5 with 3 left over. Doubled, **66 ÷ 12** is 5 with 6 left over. Both leftovers are half of the divisor.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did scaling both numbers keep in a division?** (The quotient, the whole-number part.)
- **What did it change?** (The leftover, which scaled along with the divisor.)

WHAT MAKES THIS TRUSTWORTHY?

The trust comes from knowing what the new question keeps and what it changes. Scaling both keeps the quotient for division, doubles the answer for subtraction, and changes the leftover. And, as always, the answer should be taken back to the original: if 72 ÷ 3 = 24, then 15 × 24 should be 360.

PRACTICE

1. Find 450 ÷ 15 by dividing both numbers by 5.
2. Find 2,400 ÷ 50 by dividing both numbers by 10.
3. Find 900 ÷ 36 by dividing both numbers by 9.
4. What are 17 ÷ 5 and 51 ÷ 15? Compare the quotients and the leftovers.
5. 100 − 40 = 60. Predict 300 − 120 without doing the subtraction, then check.

BRIDGE *Bridge to the rest of the series:* In chapter 3, sliding both numbers kept a difference. Here scaling both keeps a quotient. In each, the rule depends on what the question asks, and the check is to take the answer home.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × 2 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × 2 )  →  4
6 pop ( 1,600 − 4 )  →  1,596
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × 1 ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × 1 )  →  2
6 pop ( 625 − 2 )  →  623
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × ( 4 + 1 ) ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × ( 4 + 1 ) )  →  20
6 pop ( 2,500 − 20 )  →  2,480
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 5

Count the Other Side

STORY

Pia was asked: *how many of the numbers from 1 to 20 are NOT multiples of 3?*
She could list the numbers that are not multiples and count them. But she changed the question.

STEP 1 Count what you don't want

Instead of counting the ones that are not multiples of 3, Pia counted the ones that are.

TRACE

```
3, 6, 9, 12, 15, 18
```

That is 6 numbers. There are 20 numbers in all, and every number is either a multiple of 3 or not. So:

TRACE

```
( 20 − 6 )  →  14
```

There are **14** numbers that are not multiples of 3.

The question "how many are not multiples?" became "how many are multiples?", and then a subtraction from the total. The second question was easier because multiples of 3 are quick to list.

STEP 2 When it saves real work

How many numbers from 1 to 50 do NOT end in 0 or 5?

Listing the ones that do not end in 0 or 5 means writing out 40 numbers. Counting the ones that do is quicker:

TRACE

```
5, 10, 15, 20, 25, 30, 35, 40, 45, 50
```

That is 10 numbers. So (50 − 10) → **40**.

Three coins are flipped. In how many of the possible results is there at least one head?

Trust check. The reason is that everything is on one side or the other, and nothing is on both.

The results are listed by writing H or T for each coin, and there are 8 in all. Listing results with at least one head is a long job. But there is only one result with no heads at all: T T T.

TRACE

```
( 8 − 1 )  →  7
```

There are **7** results with at least one head.

STEP 3 What the change keeps, and when it fails

Pia asked her usual two questions.

- **What does the new question keep?** The answer. Every number is on one side or the other, so the two counts always add to the total.
- **What might it lose?** It only works if you know the total and if each item is on exactly one side. If some numbers were both multiples and not multiples, or neither, the subtraction would be wrong.

The change is good when "the other side" is **short** or **easy to recognise**. It is a poor change when the other side is just as long as the first.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did counting the other side keep?** (The final answer.)
- **What made it easier?** (The other side was shorter and easy to list.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is that everything is on one side or the other, and nothing is on both. So the two counts add to the total exactly. The check is to list a small case both ways: from 1 to 6, the non-multiples of 3 are 1, 2, 4, 5 (that is 4), and the multiples are 3 and 6 (that is 2), and 4 + 2 = 6. ✓

PRACTICE

1. How many numbers from 1 to 30 are NOT multiples of 5?
2. How many numbers from 1 to 40 are NOT multiples

of 4?

3. Two coins are flipped. In how many of the 4 results is there at least one head?

4. How many numbers from 1 to 12 are NOT multiples of 2 or 3? (Hint: list the numbers that are multiples of 2 or of 3.)

5. Why would counting the other side be a bad idea for "how many numbers from 1 to 20 are multiples of 3"?

BRIDGE *Bridge to the rest of the series:* In chapter 4, scaling both numbers kept a quotient. Here flipping to the other side keeps the count, as long as every item is on exactly one side. In The Order of Discovery, counting a short list is often enough when a long list would be hopeless.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **3,599**. Leave the others blank. What appears?

3601 + 8	3582 + 11	3582 + 12	3631 − 32	3624 − 25	3595 + 9	15 × 240	3569 + 26
3583 + 6	3569 + 29	3600 + 11	3638 − 39	3565 + 34	3589 + 4	3583 + 9	3582 + 11
3584 + 17	3572 + 27	3636 − 37	61 × 59	3574 + 25	3630 − 31	3635 − 36	3586 + 25
3575 + 24	3564 + 35	3591 + 7	3575 + 24	3567 + 32	3577 + 13	3577 + 22	3565 + 34
3601 − 2	3592 + 7	3616 − 17	3574 + 25	3568 + 31	3603 − 4	3595 + 4	3627 − 28
3617 − 18	61 × 59	60 × 60 − 1 × 1	3597 + 2	3633 − 34	3636 − 37	3580 + 19	3607 − 8
3589 + 18	3572 + 27	3571 + 28	3596 + 15	3599 + 5	60 × 60 − 1 × 1	3576 + 23	3568 + 27
3590 + 14	3615 − 16	3562 + 37	3602 + 4	3567 + 22	3573 + 26	3571 + 28	3581 + 9



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

SOLVING BY CHANGING THE QUESTION

CHAPTER 6

Halve One, Double the Other

STORY

Pia was asked: **16 × 25**. She did not like it. So she changed the question.

STEP 1 Halve one, double the other

Pia halved 16 and doubled 25.

TRACE

```
( 16 × 25 )
redraw  →  ( 8 × 50 )
```

She did it again.

TRACE

```
( 8 × 50 )
redraw  →  ( 4 × 100 )
pop  →  400
```

And 4 × 100 is easy. The answer is **400**. Check: 16 × 25 is 16 quarters of 100, which is 4 hundreds. ✓

Another: **35 × 12**.

TRACE

```
( 35 × 12 )
redraw  →  ( 70 × 6 )
redraw  →  ( 140 × 3 )
pop  →  420
```

Check: 35 × 12 = 35 × 10 + 35 × 2 = 350 + 70 = 420. ✓

STEP 2 Why the product does not change

Think of 16 × 25 as 16 rows of 25 chairs. Join the rows in pairs. Two rows of 25 chairs make one row of 50 chairs, so the 16 rows become 8 double rows.

TRACE

```
16 rows of 25  =  8 rows of 50
```

Trust check. The reason is the chairs: fewer, longer rows with the same chairs.

The chairs did not change. They were only arranged in longer, fewer rows. So halving one number and doubling the other keeps the total.

STEP 3 What to do with an odd number

Halving only works neatly when the number is even. **35 × 12** has an odd number, 35, so Pia halved the other one: she halved 12 and doubled 35. That keeps everything whole.

What about two odd numbers, like **15 × 17**? Neither can be halved. This trick does not help, and Pia uses another one from the earlier chapters.

STEP 4 What the change keeps and loses

- **Keeps:** the product. Every step can be undone by halving and doubling the other way, so no answers are added or lost, just as in chapter 2.
- **Loses:** the original numbers. 4 × 100 no longer mentions 16 or 25.
- **Costs:** you must be able to halve an even number and double the other cleanly.

The change is good when it ends in a friendly number like 10, 100 or 1,000. It is a poor choice when it makes the numbers harder.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What stayed the same?** (The product.)
- **What was lost?** (The original numbers.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is the chairs: fewer, longer rows with the same chairs. The change can be undone by doubling and halving back, and a change that can be undone keeps exactly the same answer.
The check is a second route, such as splitting one number, as Pia did with 35 × 12.

PRACTICE

1. Find 45 × 14 by halving 14 and doubling 45.
2. Find 125 × 8 by halving and doubling until you reach a friendly number.

3. Find 24×15.

4. Find 48×25. (Hint: halve 48 twice.)

5. Why can Pia not use this trick directly on 15×17?

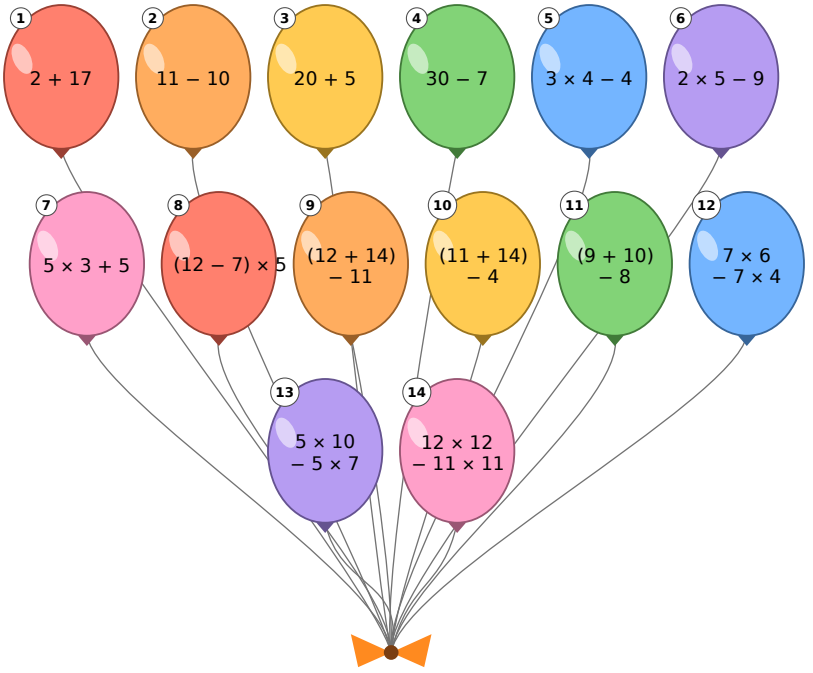
BRIDGE *Bridge to the rest of the series:* In chapter 4, scaling both numbers kept a quotient. Here halving one and doubling the other keeps a product. In The Work That Vanishes, moving one number up and the other down keeps a sum, and the corner is the price for a product.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

CHAPTER 7

Try the Small Case First

STORY

Pia was asked for **111 × 111**. She did not want to do it. So she tried a smaller question first.

STEP 1 Start small

TRACE

```
( 11 × 11 )  →  121
```

That is easy. She went a little bigger.

TRACE

```
( 111 × 111 )  →  12,321
```

She had to work that one out, but it was not too hard. And she noticed that 121 and 12,321 both read the same forwards and backwards, and the digits climb 1, 2, 3 and then fall again.

She tried one more.

TRACE

```
( 1,111 × 1,111 )  →  1,234,321
```

The same pattern. Climb to 4, fall back. So she guessed:

A number made of k ones, times itself, climbs from 1 to k and then falls back to 1.

STEP 2 Use the pattern

The pattern says 11,111 × 11,111 should climb to 5 and fall.

TRACE

```
11,111 × 11,111  →  123,454,321
```

And it does. Pia's small cases led her to the answer for a big case without doing the long multiplication.

That is the idea of this chapter: **change a big question into a small one, solve the small one, and look for a pattern.**

Trust check. A small case is trustworthy for what it shows: that the small answer is right.

STEP 3 But does the pattern last?

Pia kept going. Nine ones squared gives 12,345,678,987,654,321. The pattern holds.

Now ten ones. The pattern says it should climb to 10, but 10 is two digits. What really happens?

TRACE

```
1,111,111,111 × 1,111,111,111  →  1,234,567,900,987,654,321
```

The pattern broke. The answer starts 1,234,567, but then the middle turns into 9, 0, 0, 9 instead of climbing neatly to 10, because the "10" cannot fit in one place and spills into the next.

So the pattern held for 1 up to 9 ones, and then failed at 10. It is the same lesson as the prime recipe in The Order of Discovery: a pattern from small cases is a suggestion, not a proof.

STEP 4 Small cases are for finding, not for proving

Small cases are good for three things.

- They help you **see** a pattern.
- They help you **guess** the answer to the big case.
- They let you **check** a big answer: it should look like the small ones.

They are not a proof. A proof needs a reason, and the reason for this pattern is carrying, which is why it breaks at ten.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the small cases keep?** (The shape of the problem.)
- **What did they lose?** (Anything that only shows up at larger sizes, like carrying.)

WHAT MAKES THIS TRUSTWORTHY?

A small case is trustworthy for what it shows: that the small answer is right. The leap to the big case is a guess until it has a reason. Pia marks it **only seen so far**, and then checks the big case directly where she can.

PRACTICE

1. Find 1,111 × 1,111 by the pattern. Check it is 1,234,321.
2. Use the pattern to guess 111,111 × 111,111.
3. Check 111 × 111 by splitting: (111 × 100) + (111 × 10) + (111 × 1).
4. Why does the pattern need to break at ten ones?
5. Try the small case first for 99 × 99, 999 × 999. What do you notice?

BRIDGE *Bridge to the rest of the series:* In chapter 6, a question was made friendlier by changing both numbers. Here it is made smaller. In The Order of Discovery, a pattern that works for small cases but breaks later is labelled “only seen so far”.
Next in this book: Chapter 8: From 48 × 52 to 50 × 50.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × 3 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × 3 )  →  9
6 pop ( 1,600 − 9 )  →  1,591
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × 1 ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × 1 )  →  2
6 pop ( 400 − 2 )  →  398
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × ( 6 + 1 ) ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × ( 6 + 1 ) )  →  42
6 pop ( 4,900 − 42 )  →  4,858
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 8

From 48 × 52 to 50 × 50

STORY

Pia was given the case file card **48 × 52**. She did not answer it. She answered a *different* question, **50 × 50**, and then paid for the change.
This chapter is about what a change of question costs.

STEP 1 The slide and its price

Pia slid 48 up by 2 and 52 down by 2. The two numbers still add to 100.

TRACE

```
( 48 × 52 )   →   ( 50 × 50 )
```

The new question has a different answer. 50 × 50 = 2,500, and 48 × 52 = 2,496. The change **added 4** to the answer. That is its price.

Pia can name the price before she finishes, because she knows how the numbers moved: each is 2 from the middle, so the price is 2 × 2.

TRACE

```
( 48 × 52 )  =  ( ( 50 × 50 ) − ( 2 × 2 ) )
```

Change the question, solve the easier one, then take back the price. That is the whole trick.

STEP 2 A change whose price is hard

Pia tried a different slide: move *both* numbers up by 2.

TRACE

```
( 48 × 52 )   →   ( 50 × 54 )
```

This is not any easier. 50 × 54 = 2,700, and the price is 2,700 − 2,496 = **204**. She can name it as well: 2 × 52 + 48 × 2 + 4. But the price is not a tidy square, so paying it is nearly as much work as the original.

Trust check. Pia can say how much the answer changed because she knows how the numbers moved.

Change	New question	Price	Easy to pay?
Slide opposite by 2	50 × 50	4	Yes, 2 × 2
Slide both up by 2	50 × 54	204	No

A good change has a price that is **small** and **easy to name**.

STEP 3 Every change has a price

Pia made a rule for her notebook: *before you change a question, ask what the change costs, and write it down.*

TRACE

```
change   ( 48 × 52 )  →  ( 50 × 50 )
price    4  (the answer to the new question is 4 too high)
```

This is the same as an estimate and its debt in A Useful Wrong Answer. The only difference is that here Pia pays the debt at once, so the answer to the original question is exact.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the change keep?** (The sum of the two numbers, 100, and the shape of the problem.)
- **What did it lose?** (The exact answer to the original question, until the price was paid.)

WHAT MAKES THIS TRUSTWORTHY?

Pia can say how much the answer changed because she knows how the numbers moved. The price is a calculation, not a hope. After she pays it, she checks the result by another route, such as 48 × 50 + 48 × 2 = 2,400 + 96 = 2,496.

PRACTICE

1. Change **49 × 51** to 50 × 50. What is the price? What is the answer to 49 × 51?
2. Change **38 × 42** to 40 × 40. What is the price? Find 38 × 42.
3. Change **23 × 27** to 25 × 25. Find 23 × 27.
4. Change **48 × 52** to 48 × 50 (only the second number moves). What is the price? Use it to find 48 × 52.

5. Pia says, “I changed the question, so my answer is wrong.” Is she right?

BRIDGE *Bridge to the rest of the series:* In A Useful Wrong Answer, the price of the change is the error on an estimate. In The Work That Vanishes, the price is the corner. Here it is called what it is: the cost of changing the question.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,475**. Leave the others blank. What appears?

2475 + 8	2 × 1238	2460 + 5	2455 + 20	2437 + 38	2457 + 17	2475 + 12	2459 + 19
31 × 80	2478 + 7	2463 + 2	2468 + 7	55 × 45	2448 + 17	2468 + 19	2457 + 22
2453 + 15	2449 + 26	2450 + 25	2490 − 15	2485 − 10	2495 − 20	2452 + 23	2447 + 27
2456 + 19	2512 − 37	2482 + 2	2448 + 27	2468 + 7	2469 + 10	2466 + 9	2498 − 23
2438 + 37	2465 + 10	55 × 45	2491 − 16	2452 + 23	2463 + 12	2481 − 6	2459 + 16
2447 + 28	2463 + 12	2473 + 2	2478 − 3	2451 + 24	2506 − 31	2451 + 24	2494 − 19
2464 + 3	2456 + 19	45 × 55	2466 + 15	2454 + 9	2473 + 2	2482 − 7	2451 + 29
2477 + 7	2444 + 31	2447 + 28	58 × 58 − 30 × 30	67 × 37	33 × 75	2446 + 29	2448 + 29



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

Answers (turn the book around)

Chapter 1

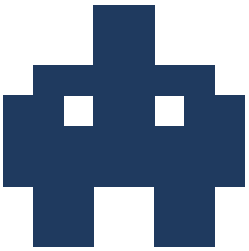
1. 995 is 5 below 1,000 and 1,005 is 5 above, so the distance is 5 + 5 = **10**.
2. 38 to 40 is 2, 40 to 70 is 30, 70 to 72 is 2. Total: **34**.
3. 12 is 10 + 2 and 15 is 10 + 5. The pieces are 10 × 10 = 100, 10 × 5 = 50, 2 × 10 = 20 and 2 × 5 = 10. Total: **180**.
4. The gap is (x + 9) − 21. At x = 10 the gap is −2, so x is too small. At x = 12 the gap is **0**, so **x = 12**.
5. It loses the whole-cookie rule. Each kid gets **2** cookies, and **2** are left over.
6. It lost the direction, meaning which number is bigger. The distance is the size of the difference without the sign.
7. ((50 − 48) + (52 − 50)) pops to (2 + 2), which pops to **4**.

Spot the Slip.
Notebook A: **perfect**. 37 × 43 = 1,591.
Notebook B: slip on line **3**. The corner is 1 × 1, not 1 × (1 + 1). The answer should be 399.
Notebook C: slip on line **5**. 6 × 6 is 36. The answer should be 4,864.

Chapter 2

1. Adding 3 gives x + 5 = 12. Then x = 7. Yes, you can undo it by subtracting 3, so the answers are the same: **x = 7**.
2. The new question is 0 = 0, which is true for **every** number. Only x = 7 is an answer to the original.
3. Both x = 0 and x = 4 work: 0 × 6 = 0 × 6 and 4 × 6 = 4 × 6. Every number works, since both sides are the same.
4. She gets 2 = 7, which is false. She lost x = 0, since 0 × 2 = 0 and 0 × 7 = 0. So **x = 0** is an answer.
5. The doubling button, because it can be undone by halving. The ×0 button cannot be undone, so it might add answers.

Color by Answer (2,496). The shaded boxes make a robot.



Answers (turn the book around)

Chapter 3

1. (502 − 297) → (505 − 300) → **205**.
2. (93 − 50) pops to **43**. Check: 43 + 48 = 91. ✓
3. Slide both up by 2: (6,003 − 3,000) → **3,003**.
4. Move 4 from 127 to 396: (400 + 123) → **523**.
5. Sliding both numbers adds the slide twice to a sum. 52 + 38 = 90, but 54 + 40 = 94. For a sum, move one up and the other down: (54 + 36) → 90.
6. Slide both up by 1: (72 − 40) → **32**.

Balloon Message.

1.	3 + 11	= 14	N
2.	24 − 23	= 1	A
3.	25 − 12	= 13	M
4.	7 − 2	= 5	E
5.	5 × 4	= 20	T
6.	4 × 2	= 8	H
7.	2 × 7 − 9	= 5	E
8.	(15 + 9) − 3	= 21	U
9.	7 × (11 − 9)	= 14	N
10.	(5 + 14) − 8	= 11	K
11.	(12 + 6) − 4	= 14	N
12.	8 × 8 − 7 × 7	= 15	O
13.	12 × 12 − 11 × 11	= 23	W
14.	2 × 12 − 2 × 5	= 14	N

Message: **NAME THE UNKNOWN!**

Chapter 4

1. (90 ÷ 3) pops to **30**. Check: 15 × 30 = 450. ✓
2. (240 ÷ 5) pops to **48**. Check: 50 × 48 = 2,400. ✓
3. (100 ÷ 4) pops to **25**. Check: 36 × 25 = 900. ✓
4. 17 ÷ 5 is 3 with 2 left over. 51 ÷ 15 is 3 with 6 left over. The quotients agree. The leftover tripled, as the divisor did.
5. Both numbers are three times as big, so the answer is **180**. Check: 300 − 120 = 180. ✓

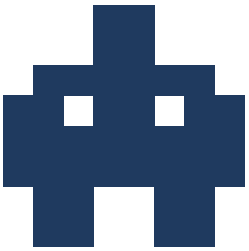
Spot the Slip.
Notebook A: **perfect**. 38 × 42 = 1,596.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 624.
Notebook C: slip on line **3**. The corner is 4 × 4, not 4 × (4 + 1). The answer should be 2,484.

Answers (turn the book around)

Chapter 5

1. Multiples of 5 up to 30: 5, 10, 15, 20, 25, 30, which is 6. (30 − 6) → **24**.
2. Multiples of 4 up to 40: 4, 8, ..., 40, which is 10. (40 − 10) → **30**.
3. The results are HH, HT, TH, TT. Only TT has no head. (4 − 1) → **3**.
4. Multiples of 2 or 3 up to 12: 2, 3, 4, 6, 8, 9, 10, 12, which is 8. (12 − 8) → **4**. (They are 1, 5, 7 and 11.)
5. You would count the 14 non-multiples first and then subtract, which is longer than just listing the 6 multiples.

Color by Answer (3,599). The shaded boxes make a robot.



Chapter 6

1. (90 × 7) pops to **630**.
2. (250 × 4) → (500 × 2) → **1,000**.
3. (12 × 30) pops to **360**.
4. (24 × 50) → (12 × 100) → **1,200**.
5. Both numbers are odd, so neither can be halved into a whole number. She needs a different move.

Balloon Message.

1.	2 + 17	= 19	S
2.	11 − 10	= 1	A
3.	20 + 5	= 25	Y
4.	30 − 7	= 23	W
5.	3 × 4 − 4	= 8	H
6.	2 × 5 − 9	= 1	A
7.	5 × 3 + 5	= 20	T
8.	(12 − 7) × 5	= 25	Y
9.	(12 + 14) − 11	= 15	O
10.	(11 + 14) − 4	= 21	U
11.	(9 + 10) − 8	= 11	K
12.	7 × 6 − 7 × 4	= 14	N
13.	5 × 10 − 5 × 7	= 15	O
14.	12 × 12 − 11 × 11	= 23	W

Message: **SAY WHAT YOU KNOW!**

Answers (turn the book around)

Chapter 7

1. Climb to 4 and fall: **1,234,321**.
2. Climb to 6 and fall: **12,345,654,321**.
3. 11,100 + 1,110 + 111 = **12,321**. ✓
4. The digit would have to be 10, which does not fit in one place, so it carries into the next place and changes the pattern.
5. 99 × 99 = 9,801 and 999 × 999 = 998,001. Each is a round number minus a small number: 10,000 − 199 and 1,000,000 − 1,999. Another pattern, with a reason: it is the square-with-a-step idea from The Work That Vanishes.

Spot the Slip.
Notebook A: **perfect**. 37 × 43 = 1,591.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 399.
Notebook C: slip on line **3**. The corner is 6 × 6, not 6 × (6 + 1). The answer should be 4,864.

Chapter 8

1. Each is 1 from the middle, so the price is 1 × 1 = **1**. 2,500 − 1 = **2,499**.
2. Each is 2 from the middle, so the price is **4**. 1,600 − 4 = **1,596**.
3. Each is 2 from the middle, so the price is 4. 625 − 4 = **621**.
4. 48 × 50 = 2,400 is too low by 48 × 2 = **96**. 2,400 + 96 = **2,496**.
5. Only if she forgets to pay the price. A changed question gives a different answer, but a named price turns it back into the answer to the original question.

Color by Answer (2,475). The shaded boxes make a robot.

